DATA VORTEX — ROUND 1 (PHASE 2)

LOGIC EXPLANATION

Theme: Rebuilding the Social Engine

Participant details:

Team Name: Joy

Team Size: 1

Team Head: Thathi Reddy Jivan Reddy

Selected Questions: E3, M5, H4

Database: MySQL 8.0.46

Schema: data_vortex

Tables Used: posts.users

DATABASE & OVERALL APPROACH

The analysis was performed using MySQL 8.0.46 on the data_vortex schema, using the posts and users tables.
The selected questions are E3, M5, and H4.

The SQL logic uses dynamic calculations, aggregation, filtering, joins, and window functions rather than hardcoded analytical outputs. This keeps the results dependent on the underlying dataset and makes the analysis reproducible.

Missing values are handled explicitly wherever required by the individual question. In particular, platform-level

analysis ignores missing platform values, while engagement comparisons exclude incomplete engagement records where the question requires complete values.

1. E3 - AVERAGE ENGAGEMENT BY PLATFORM

Question: Calculate the average likes, shares, and comments for each platform and identify the platform with the highest average total engagement.

Logic:

1. Filter out **NULL** and blank platform values because the question requires missing platform values to be ignored.
2. Group posts by platform using **GROUP BY.**
3. Calculate average likes, shares, and comments using **AVG()**.
4. Because likes is stored as text in the imported MySQL table, convert it to a numeric value with **CAST(likes AS DECIMAL(10,2))** before numeric calculations.
5. Calculate total engagement as **likes + shares + comments.**
6. Calculate average total engagement only when likes, shares, and comments are all available.
7. Use **ORDER BY avg_total_engagement DESC** so the highest-engagement platform appears first.

Result: *Instagram ranked first with an average total engagement of 3,669.38.*

Why this query is useful: It provides a platform-level comparison and identifies which social platform generates the strongest average engagement.

2. M5 - DETECT SUSPICIOUS ENGAGEMENT

Question: Identify posts where shares are greater than likes plus comments and return the top 20 by total engagement.

Logic:

1. Exclude posts where likes, shares, or comments are **NULL** so that the comparison is based on complete engagement data.
2. Apply the suspicious engagement condition: **shares > likes + comments.**
3. Convert likes from text to **DECIMAL** using **CAST()**.
4. Calculate total engagement as **likes + shares + comments.**
5. Sort qualifying posts using **ORDER BY total_engagement DESC.**

6. Use **LIMIT 20** to return the top 20 highest-impact posts satisfying the suspicious engagement condition.

Result: *20 posts were returned.*
The highest-total-engagement qualifying post had total engagement of 3,936.

Why this query is useful: It identifies posts where sharing activity is unusually high compared with likes and comments, which can indicate unusual engagement behaviour requiring further investigation.

3. H4 - FOLLOWER TO ENGAGEMENT ANOMALY

Question: Identify users with fewer than 5,000 followers whose total post engagement falls within the top 10% of all users.

Logic:

1. Join the users and posts tables using **user_id.**
2. Use a **LEFT JOIN** so users remain represented even when post activity is absent.

3. Aggregate the data at user level using **user_id, location,** and **follower_count**.

4. Calculate each user's total engagement from posts where likes, shares, and comments are available.

5. Use a Common Table Expression (CTE) named **user_engagement** to create the user-level engagement dataset.

6. Use a second CTE named **ranked_users**.

7. Apply the **CUME_DIST()** window function to rank users according to total engagement in descending order.

8. Select users whose engagement percentile is <= **0.10**, representing the top 10% engagement group.

9. Apply the anomaly condition: **follower_count** < **5000**.

10. Sort the final results by total engagement in descending order.

Result: 15 users satisfied both conditions.
Highest qualifying user: User ID user_uerv85na •Rome, Italy
• Followers 1,824
•Total Engagement 64,753

Why this query is useful: It identifies users who have relatively small audiences but generate exceptionally high engagement, revealing potential follower-to-engagement

anomalies.

SQL TECHNIQUES USED

Question	**SQL techniques**
E3	•GROUP BY •AVG() •COUNT() •CASE •CAST() •ROUND() •ORDER BY
M5	•WHERE conditions •CAST() •Calculated field • ORDER BY •LIMIT
H4	•CTE •LEFT JOIN •GROUP BY •CASE •SUM() •CUME_DIST() •Percentile filtering •ORDER BY

OVERALL ANALYTICAL APPROACH

- **E3** → Descriptive platform-level analysis
- **M5** → Post-level suspicious engagement detection
- **H4** → User-level multi-table anomaly analysis

VALIDATION

All three SQL queries were executed successfully in **MySQL Workbench** using **MySQL 8.0.46**.